Krish Nagaral

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Skills

AI & Machine Learning: Model Training, Fine Tuning, LoRA, Transfer Learning, Generative AI, Large Language Models, RAG Pipelines, CNN, RNN, Reinforcement Learning, Langchain, Reinforcement Learning
Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, XGBoost, Light- GBM
MLOps & Deployment: MLflow, Model Versioning, Experiment Tracking, Feature Pipelines, Batch Inference, Real Time Inference, Model Monitoring, Grpc, Prometheus, Grafana, Azure AI Foundry, Copilot Studio
Cloud & DevOps: AWS(SageMaker, EC2, ECS, EKS, Lambda, S3), Docker, Kubernetes, CI/CD, CloudWatch, AKS
Programming: Python, Java, C++, SQL, JavaScript, FastAPI, Flask, REST APIs, Websockets

Experience

AI Research & Tool Developer Intern, Coutloot & Flyads.ai - Mumbai — Aug 2024 – Aug 2025
- Accomplished 92 percent accuracy improvement by fine tuning LLaMA and Gemma models using LoRA
- Accomplished scalable inference by deploying Docker based FastAPI services supporting 5000 plus concurrent requests
- Accomplished latency reduction by optimizing batching strategies and GPU utilization

Software Tester Intern, Coutloot - Mumbai — Apr 2024 – Jun 2024
- Designed testing frameworks, improving application stability by 88% with 99.8% product reliability before deployment
- Recorded and reported test cases, ensuring traceability and actionable insights for developers
- Improved application performance by systematically identifying and resolving bugs

Cloud Engineer Intern, Tata Communications - Mumbai — Sep 2023 – Mar 2024
- Authored deployment documentation and training materials to streamline onboarding
- Supported live systems by troubleshooting incidents and collaborating with cross-functional teams
- Applied best practices in AWS EC2, S3, and Lambda to deliver resilient cloud solutions

Projects

CodeMentor AI: Multi-Agent Codebase Assistant (live link) | FastAPI, Next.js, RAG, LangGraph, SSE
- Orchestrated 5 LangGraph agents (planner, retrieval, analyst, mentor, memory) behind interface-driven DI across 11 FastAPI endpoints, pairing FAISS vector search with Tree-sitter AST parsing to answer codebase queries with p95 latency under 800ms over streaming SSE.
- Instrumented the full stack with Prometheus + Grafana tracking retrieval precision, latency percentiles, and utilization, then ran 3 data-driven tuning cycles that measurably improved answer relevance; containerized with Docker, deployed on Render/Vercel, and gated merges with GitHub Actions

BoxBox: F1 Pit Stop Strategy Optimizer| PPO, Gymnasium, gRPC, Flask, Prometheus, Grafana, Docker, Kubernetes
- Implemented a custom Gymnasium environment simulating F1 races lap-by-lap from 45,623 real telemetry laps across 42 Grand Prix, trained a PPO agent that gains 5.7 positions over a fixed-window baseline (P7.8 vs P13.5 mean finish) with a 1.3-stop strategy matching real F1 race patterns.
- Served the RL agent over gRPC hitting 4.67ms p95 inference (40x under the 200ms target), streamed live recommendations via Flask-SocketIO, and monitored action distribution, latency percentiles, and model confidence through a 5-panel Grafana dashboard backed by Prometheus.

Medical Imaging AI System for Pneumonia Detection
- Built ResNet based deep learning classifier on 5800 plus chest X ray images achieving 91.5 percent accuracy and AUROC of 0.98
- Implemented Grad CAM explainability for model interpretability enabling clinical validation and trust
- Deployed real time inference APIs using FastAPI and Docker with sub 200ms response latency

Credit Risk Scoring System with Explainable AI
- Built XGBoost based credit risk prediction model on 200000 plus financial records achieving 91 percent accuracy
- Implemented SHAP based explainability enabling 95 percent model interpretability for regulatory compliance
- Deployed model serving infrastructure using Flask API with automated feature pipelines using Airflow

Publications

Comparative Analysis of Machine Learning Algorithms for Heart Attack Prediction – IJFMR, 2024
Quantum Cryptography: Quantum Key Distribution Protocols for Secure Communication – IJNRD
Green and Speculative Computing: Energy Efficient Computing Strategies – IJNRD

Education

Northeastern University – MS Computer Science, AI & ML Specialization — Sep 2025 – May 2027
Vellore Institute of Technology – BTech Computer Science — Aug 2021 – May 2025

Certifications

AWS Solution Architect Associate | AWS AI Practitioner | AWS Machine Learning Associate | Deep learning.ai Prompt Engineering | Deep learning.ai Machine learning | Hugginface MCP | Nvidia RAG | IBM Cybersecurity